

Simulation and Detection of SSH-Based Botnet Attacks

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Abstract—The Mirai malware is known for some of the largest distributed denial-of-service (DDoS) attacks. It infects Linux-based devices and turns them into remotely controlled bots for large-scale network attacks. This project simulates the full lifecycle of an SSH-based botnet attack — reconnaissance, credential brute-forcing, lateral movement through segmented networks, and command-and-control communication — in a controlled, containerised environment. A key contribution is a real-time browser dashboard that visualises the live attack topology via Server-Sent Events, integrates scenario setup, and enables live application of defenses so their effect can be observed as the botnet runs. Four defensive countermeasures are evaluated — Fail2Ban, firewall IP blocking, SSH rate limiting, and disabling password authentication — across four network scenarios of increasing complexity, demonstrating that layered defenses are necessary to prevent multi-stage compromise and lateral movement. All code, configuration, and instructions to reproduce the experiments are at <https://github.com/wendellgust/ssh-botnet-lab>.

I. INTRODUCTION

Context. Cyberattacks increasingly involve multi-stage techniques combining information gathering, credential attacks, lateral movement, and command-and-control (C2) communication. The SSH protocol, while secure by design, is frequently misconfigured in production environments — particularly through enabled password authentication — making it a common entry point for automated botnet propagation, as demonstrated by malware families such as Mirai [1], [2].

Problem. Traditional cybersecurity courses focus on isolated attack techniques without demonstrating the complete attack lifecycle. Real-world intrusions involve sequential phases, and the interaction between those phases and defensive countermeasures is not well understood in educational settings.

Goals. This project addresses three questions:

- 1) Can a Python-based botnet autonomously propagate across segmented networks using only SSH brute-force and pivoting?
- 2) How effective are common SSH hardening measures (Fail2Ban, IP blocking, rate limiting, key-only auth) individually and in combination?
- 3) Does attack depth (number of pivot hops) affect the relative effectiveness of each defense?

Unlike existing SSH security exercises that test individual techniques in isolation, this project contributes an integrated environment covering the complete botnet lifecycle — initial brute-force, multi-hop lateral movement, honeypot evasion, and C2 beaconing — together with a real-time browser

dashboard that lets an operator observe the attack progressing live, apply defenses mid-run, and immediately see their effect on the network topology. The lab runs entirely in rootless Podman containers with no internet access. All code and instructions to reproduce the experiments are at <https://github.com/wendellgust/ssh-botnet-lab>.

II. RELATED WORK

The Mirai botnet became widely known after launching large-scale DDoS attacks in 2016 [1]. A comprehensive post-mortem by Antonakakis et al. [2] characterised its scanning pipeline — TCP SYN sweep, credential brute-force, and C2 enrolment — which directly inspired our botnet’s attack phases. Mirai demonstrated that weak default SSH credentials are sufficient to compromise devices at scale without exploiting any software vulnerability.

The SSH protocol was introduced by Ylönen [3] as a cryptographically secure remote shell; yet password authentication — retained for backward compatibility — remains a persistent misconfiguration in production environments [4]. Our lab reproduces this exact condition: all victim containers have password authentication enabled with weak credentials, mirroring real-world deployments.

Honeypot-based detection [5] provides high-confidence alerts because legitimate traffic never reaches a honeypot. We adopt this principle directly: our honeypot container is reachable only by traversing an internal network segment, so any connection to it is definitive evidence of successful lateral movement.

Network segmentation and defense-in-depth [6], [7] are the primary architectural countermeasures evaluated. Our multi-scenario design tests whether segmentation alone is sufficient and at what point combined controls become necessary — a question the literature addresses theoretically but rarely validates empirically in a reproducible lab environment.

III. PROJECT DESCRIPTION

A. Lab Architecture

The lab consists of Podman-based containerised hosts running Ubuntu 22.04, connected through isolated bridge networks with no internet access. Four system roles cooperate to simulate the full attack and defense lifecycle:

- **Attacker:** Python botnet that sequentially scans for live SSH hosts, brute-forces credentials from a wordlist with configurable inter-attempt delay, installs a relay on

compromised dual-homed hosts, and propagates laterally through each discovered network segment.

- **Victims:** Ubuntu containers running real OpenSSH server with intentionally weak credentials (`labuser` with common passwords). Every authentication attempt — successful or failed — is recorded in the container’s `/var/log/auth.log`, producing authentic forensic evidence.
- **Honeypot:** Mimics a legitimate internal host but logs every inbound connection with timestamp, source IP, and protocol. Because no legitimate traffic should ever reach it, any connection is a confirmed indicator of successful lateral movement.
- **GUI server:** Custom Python HTTP/SSE server (no external framework) providing a real-time browser dashboard at `http://localhost:5000`.

B. Real-Time GUI

The GUI is a central contribution of the lab. It serves a live network topology diagram rendered in SVG, updated in real time via Server-Sent Events (SSE) as the botnet progresses. Nodes change colour to reflect state transitions — idle, scanning, brute-forcing, compromised — and animated packet dots travel along edges to visualise active connections.

Beyond visualisation, the GUI integrates the full lab workflow. A scenario selector (S1–S4) triggers container setup via `auto_run.sh` and streams build output to the log feed; the *Run Botnet* button activates only after setup completes. A *Defenses* panel allows applying any of the four countermeasures to any victim with a single click, by executing the corresponding iptables, Fail2Ban, or sshd commands inside the target container via `podman exec`. This enables live, mid-run defense application — an operator can observe the attack stalling in real time as a defense takes effect. An attack report panel summarises discovered hosts, credentials found, and pivot paths at any point during or after a run.

C. Scenarios

Four scenarios of increasing complexity are defined (see Appendix A for topology diagrams):

- **S1 — Flat:** Single network (172.21.0.0/24). All hosts directly reachable.
- **S2 — Single pivot:** Two networks. Attacker must compromise Victim 1 (dual-homed) to reach `internal_net` (10.10.0.0/24).
- **S3 — Parallel pivots:** Three networks. Two independent lateral movement paths via Victim 1 and Victim 2.
- **S4 — Deep chain:** Three networks. Two-hop chain: attacker → Victim 1 → Victim 3 → `deep_net` (10.30.0.0/24).

D. Attack Phases

Each scenario executes the following phases automatically:

- 1) **Scan:** The botnet iterates over known IP ranges for the current network segment, attempting an SSH connection on port 22 to identify live hosts.

- 2) **Brute-force:** For each live host, the botnet tries all credential pairs from the wordlist via Paramiko. A configurable delay between attempts controls attack speed and influences whether rate-based defenses trigger.
- 3) **Pivot preparation:** Once a dual-homed host is compromised, the botnet opens a Paramiko `direct-tcpip` channel through the host’s existing SSH daemon — a standard SSH port-forwarding mechanism. This creates a transparent TCP tunnel to the next network segment without writing any files to the pivot host. From the perspective of internal hosts, SSH connections arrive from the pivot’s internal IP, not from the external attacker.
- 4) **Lateral movement:** The botnet tunnels new SSH connections through the relay to reach hosts on internal segments that are otherwise unreachable from the attacker. The same scan-and-brute-force logic repeats for each discovered segment.
- 5) **Honeypot trigger:** From inside the compromised network, the botnet attempts a connection to the honeypot IP. This step intentionally produces a detectable event to test IDS alerting for lateral movement.
- 6) **C2 beaconing:** Each compromised bot establishes a TCP connection to the attacker’s listener on port 8888 and sends five JSON heartbeat messages identifying itself. Bots in deeper segments relay their beacons through the chain of pivot hosts, demonstrating multi-hop C2 communication.

E. Defenses

Four countermeasures are applied to Victim 1 (the primary target and pivot host) and evaluated independently and in combination. Each defense is enforced by executing the corresponding command inside the container via `podman exec` — no container restart required.

- **Fail2Ban:** Monitors `/var/log/auth.log` and inserts a temporary iptables DROP rule for the attacker’s IP after 3 failed SSH attempts within 60 s. Reactive — requires failed attempts to trigger; the attacker may succeed before the ban fires if the first credential in the wordlist is correct.
- **Block IP:** Inserts a permanent iptables DROP rule for the attacker’s source IP (172.21.0.10). Proactive and absolute — all packets from that IP are silently dropped at the kernel level, regardless of SSH configuration. Ineffective against spoofed source IPs or lateral movement originating from a different host.
- **Rate limit:** Uses iptables `conntrack` to allow at most 10 new SSH connections per 60 s. Does not block the attacker — it slows attempts to a rate that may extend the engagement window past a practical threshold, but a sufficiently patient attacker can still brute-force successfully.
- **Disable password:** Sets `PasswordAuthentication no` in `/etc/ssh/sshd_config` and reloads `sshd`. This eliminates the brute-force vector entirely on the protected host, regardless of password strength or wordlist.

size; the only remaining attack path is public-key compromise, which is outside this threat model.

F. Replication

The full environment is reproducible from the repository. A Linux host with Podman is the only dependency. Running `bash scripts/auto_run.sh 2` builds all images, starts the containers, executes the full attack chain, and prints a detection report.

IV. ASSESSMENT

A. Methodology

All data was collected using `scripts/defense_test.sh`, an automated script that cycles through all seven defense configurations for a given scenario. For each configuration, the script executes the following workflow:

- 1) Restarts all victim containers to clear active iptables rules, Fail2Ban state, and system authentication logs;
- 2) Applies the target defensive countermeasures exclusively to Victim 1 via `podman exec`;
- 3) Triggers the autonomous attack via the GUI `/run` endpoint, executing `botnet.py` inside the attacker container;
- 4) Awaits botnet execution baseline completion to extract `/var/log/auth.log` from all victims, alongside iptables states and a structured JSON report.

TABLE I: IP assignments across all scenarios. Dashes indicate host absent. Pivot hosts are dual-homed.

Host	Network	S1	S2	S3	S4
attacker	attack_net	172.21.0.10	172.21.0.10	172.21.0.10	172.21.0.10
victim1	attack_net	172.21.0.20	172.21.0.20	172.21.0.20	172.21.0.20
	internal_net	—	10.10.0.20	10.10.0.20	10.10.0.20
victim2	attack_net	172.21.0.21	172.21.0.21	172.21.0.21	172.21.0.21
	extra_net	—	—	10.20.0.20	—
victim3	attack_net	172.21.0.22	—	—	—
	internal_net	—	10.10.0.10	10.10.0.10	10.10.0.10
	deep_net	—	—	—	10.30.0.20
honeypot	internal_net	—	10.10.0.50	10.10.0.50	10.10.0.50
victim4	extra_net	—	—	10.20.0.10	—
	deep_net	—	—	—	10.30.0.10
victim5	extra_net	—	—	10.20.0.11	—
	deep_net	—	—	—	10.30.0.11

B. Results

Table II summarises results across all four scenarios and seven configurations. Full charts are in Appendix A.

Goal 1 — Autonomous propagation. Confirmed in all baseline runs. The botnet discovered live hosts, brute-forced credentials, prepared pivot relays, and propagated to internal segments without any manual intervention. In S2 and S3 the honeypot was triggered from inside the internal network segment, confirming successful lateral movement. In S4, eight hosts were compromised across three network layers via a two-hop chain — demonstrating that the botnet correctly tunnelled its scanning and brute-force logic through two consecutive relay nodes before reaching the deepest segment.

Goal 2 — Defense effectiveness. Disabling password authentication fully eliminates brute-force against the protected host — credential attempts become impossible — but undefended hosts on the same segment remain reachable. IP blocking has the same scope limitation. Layered defense (Fail2Ban + rate limit) stops all propagation in every scenario, including S3 with its two independent pivot paths.

Goal 3 — Effect of attack depth. Depth amplifies the failure of single-control defenses. In S3 (two parallel pivots), Fail2Ban yields the same 8 compromised hosts and successful pivot as the baseline. This is a direct consequence of Fail2Ban’s reactive design: it fires only after repeated failures, so if the correct credential appears early in the wordlist, Victim 1 is compromised before the ban can trigger. Furthermore, even when the ban does fire on Victim 1, Victim 2 on `attack_net` remains entirely undefended, keeping the `extra_net` pivot path fully operational. The existence of a second undefended pivot therefore nullifies any benefit Fail2Ban provides in this topology.

IP blocking and disabling password authentication each reduce compromise to 4 hosts — half the baseline of 8 — by eliminating Victim 1’s pivot path entirely. However, Victim 2 remains directly reachable from `attack_net` and continues to function as an unimpeded pivot into `extra_net`, allowing `victim4` and `victim5` to be compromised. A successful pivot is therefore still recorded in both configurations.

Rate limiting on Victim 1 prevents its compromise but leaves Victim 2 untouched. In our experimental runs, Victim 2 also remained uncompromised due to randomised credential ordering and the fixed engagement window. This outcome is not guaranteed: a different credential shuffle would likely compromise Victim 2, granting access to `extra_net`. Rate limiting is therefore effective only when applied to all entry points — a lesson the S3 topology explicitly demonstrates.

In S4, network topology concentrates all dependency on a single sequential chain. Reaching `deep_net` requires first compromising `internal_net` through Victim 1, and subsequently using Victim 3 — dual-homed on `internal_net` and `deep_net` — as a second relay. Because every downstream host is accessible only through this chain, rate limiting at the perimeter collapses the full attack path without any need to defend internal hosts, yielding zero compromises.

C. Limitations

The credential wordlist is small and targeted; larger dictionaries or credential-stuffing lists would alter Fail2Ban thresholds. All containers run on a single physical host, which may compress attack durations. The botnet uses a fixed inter-attempt delay; adaptive scanning could evade rate-limiting. The C2 channel is unencrypted; real botnets use encrypted or peer-to-peer C2 that is harder to detect.

V. CONCLUSION

This project demonstrated that SSH-based botnet propagation across segmented networks is feasible with minimal tooling when password authentication is enabled. Four defensive

TABLE II: Compromised hosts and pivot success across all scenarios and defense configurations.

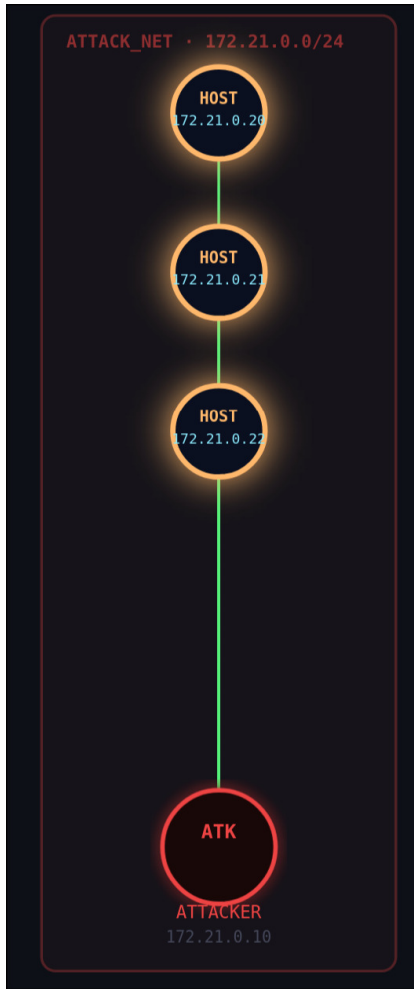
Defense	Scenario 1		Scenario 2		Scenario 3		Scenario 4	
	Comp.	Pivot	Comp.	Pivot	Comp.	Pivot	Comp.	Pivot
Baseline	3	No	4	Yes	8	Yes	8	Yes
Fail2Ban	1	No	0	No	8	Yes	1	No
Block IP	1	No	1	No	4	Yes	1	No
Rate Limit	0	No	0	No	0	No	0	No
Disable Password	1	No	1	No	4	Yes	1	No
Fail2Ban + Rate Limit	0	No	0	No	0	No	0	No
All Defenses	1	No	0	No	0	No	1	No

countermeasures were evaluated across four topologies of increasing complexity. No single defense is universally effective — disable-password configurations is the strongest standalone control, but multi-path scenarios require layered defenses. Combining Fail2Ban with rate limiting consistently stopped propagation in all scenarios. The fully containerised, reproducible lab provides a reusable platform for studying attacker behaviour and evaluating defensive mechanisms. All artifacts are available at <https://github.com/wendellgust/ssh-botnet-lab>.

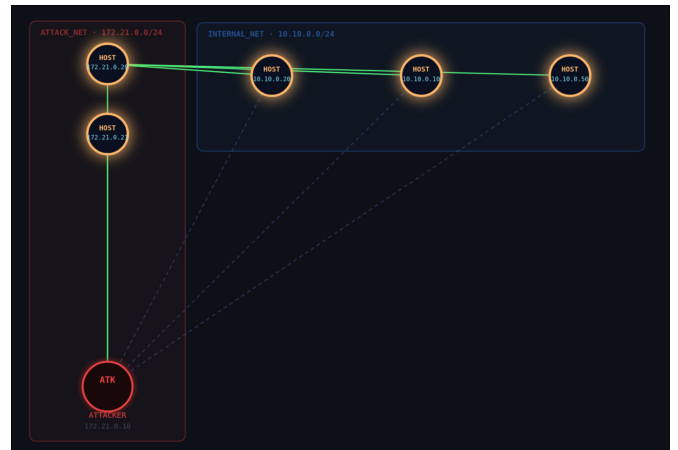
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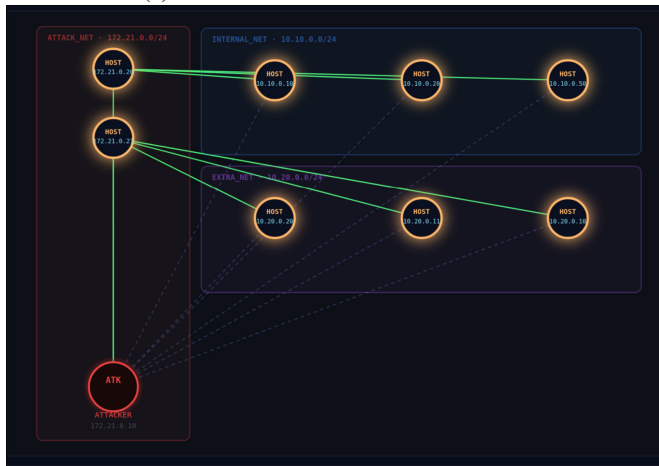
APPENDIX



(a) S1 — Flat



(b) S2 — Single pivot



(c) S3 — Parallel pivots

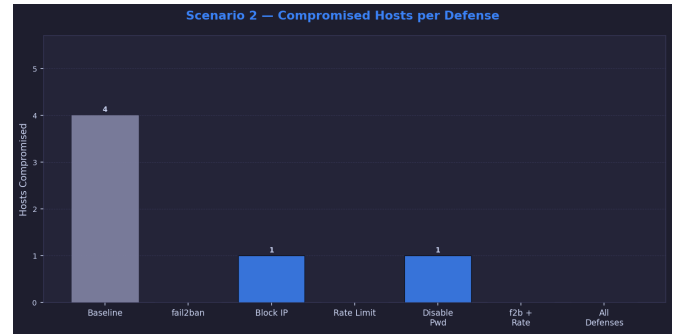


(d) S4 — Deep chain

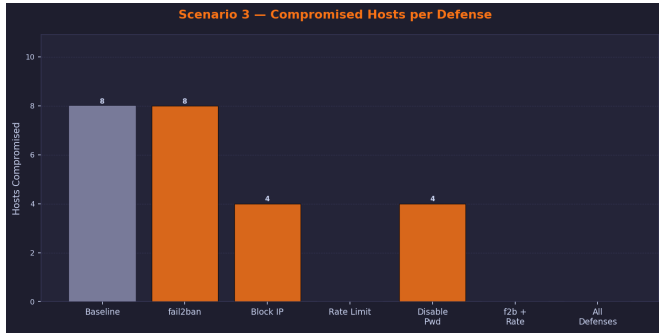
Fig. 1: Network topology for all four scenarios.



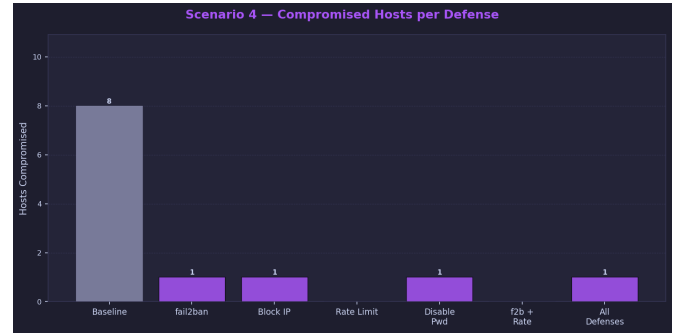
(a) S1



(b) S2

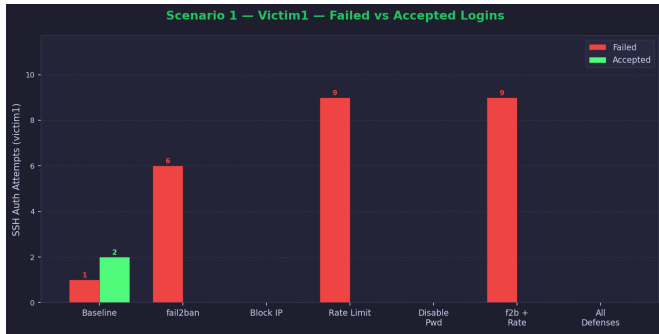


(c) S3

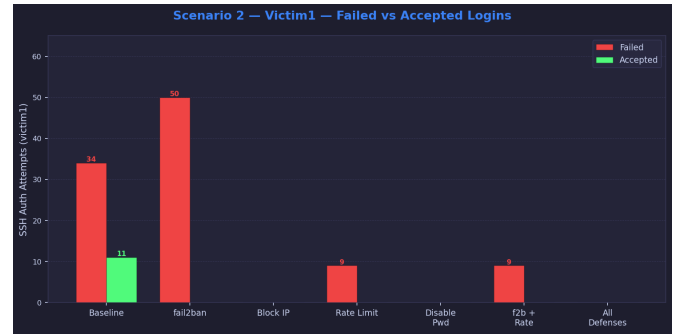


(d) S4

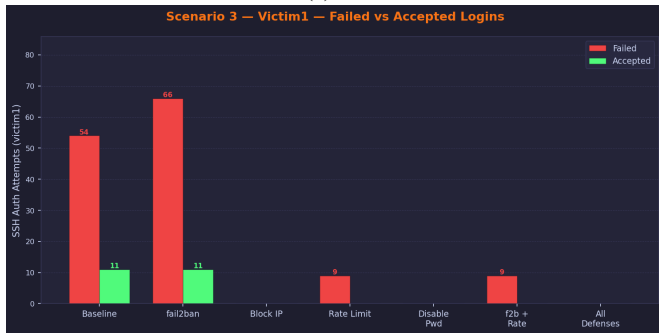
Fig. 2: Compromised hosts per defense configuration.



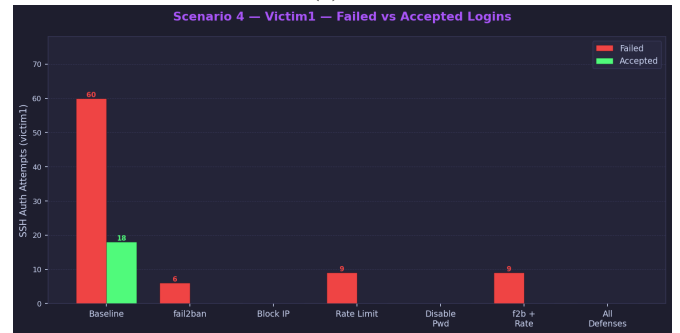
(a) S1



(b) S2

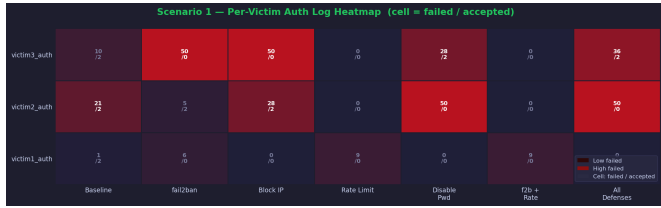


(c) S3

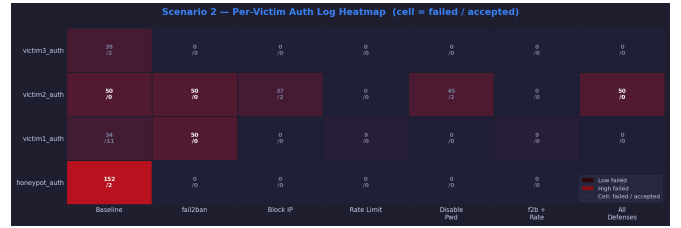


(d) S4

Fig. 3: Failed and accepted SSH authentication attempts per defense.



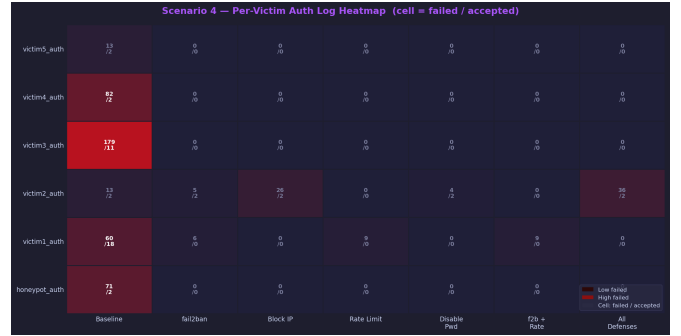
(a) S1



(b) S2

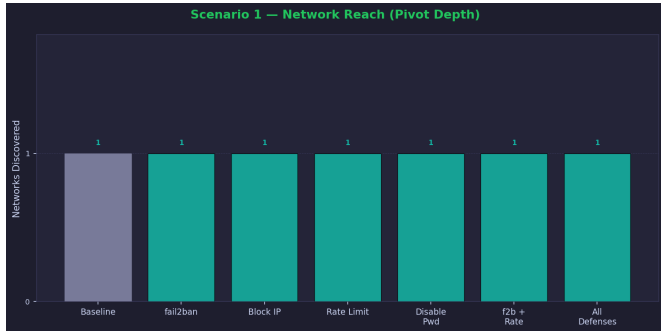


(c) S3

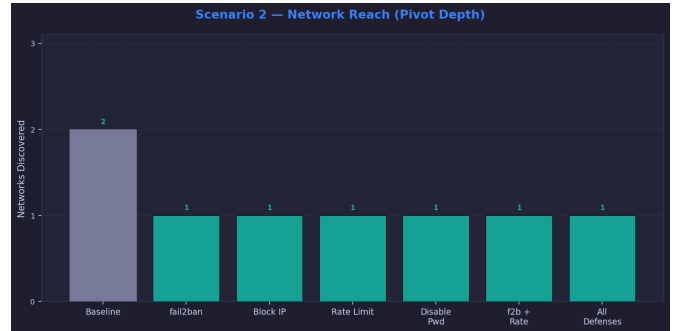


(d) S4

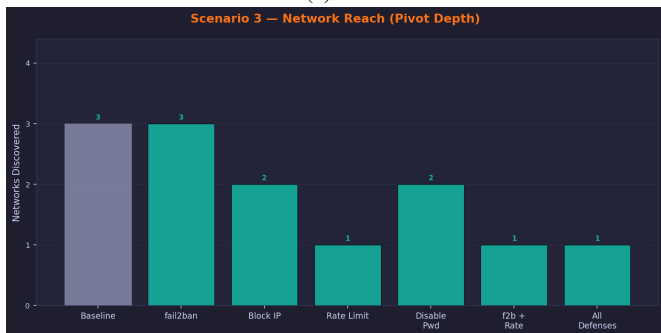
Fig. 4: Heatmap of brute-force activity across hosts and defense configurations.



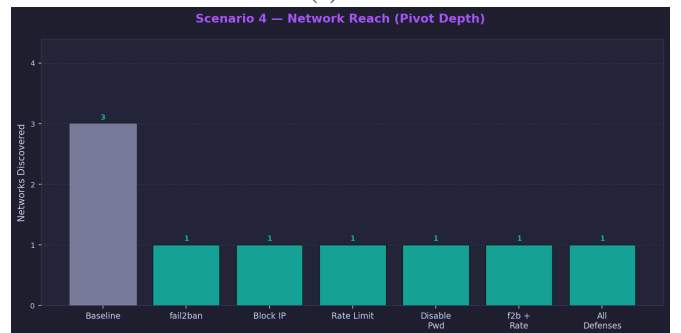
(a) S1



(b) S2



(c) S3



(d) S4

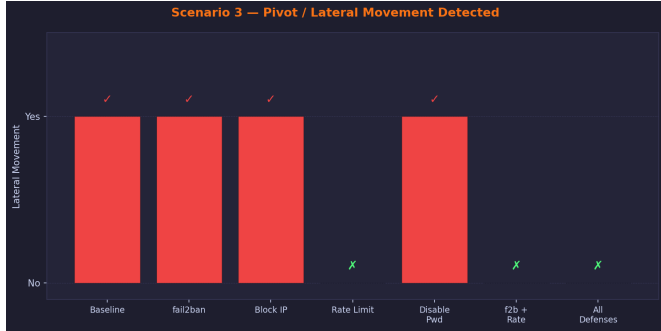
Fig. 5: Number of network segments reached per defense configuration.



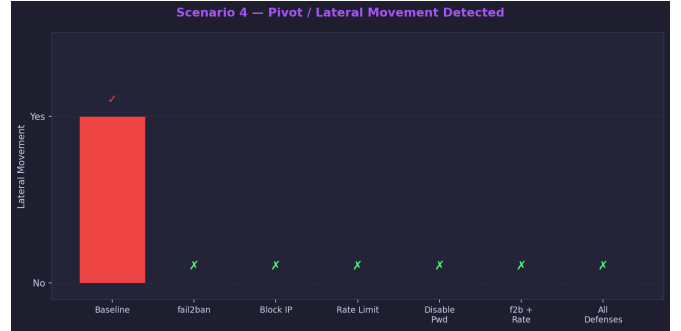
(a) S1



(b) S2



(c) S3



(d) S4

Fig. 6: Pivot success per defense configuration.

Scenario 1 — Summary Table

Defense	v1 Failed	v1 Accepted	Compromised	Networks	Pivoted
Baseline	1	2	3	1	NO
fail2ban	6	0	1	1	NO
Block IP	0	0	1	1	NO
Rate Limit	9	0	0	1	NO
Disable Pwd	0	0	1	1	NO
f2b + Rate	9	0	0	1	NO
All Defenses	0	0	1	1	NO

(a) S1

Scenario 2 — Summary Table

Defense	v1 Failed	v1 Accepted	Compromised	Networks	Pivoted
Baseline	34	11	4	2	YES
fail2ban	50	0	0	1	NO
Block IP	0	0	1	1	NO
Rate Limit	9	0	0	1	NO
Disable Pwd	0	0	1	1	NO
f2b + Rate	9	0	0	1	NO
All Defenses	0	0	0	1	NO

(b) S2

Scenario 3 — Summary Table

Defense	v1 Failed	v1 Accepted	Compromised	Networks	Pivoted
Baseline	54	11	8	3	YES
fail2ban	66	11	8	3	YES
Block IP	0	0	4	2	YES
Rate Limit	9	0	0	1	NO
Disable Pwd	0	0	4	2	YES
f2b + Rate	9	0	0	1	NO
All Defenses	0	0	0	1	NO

(c) S3

Scenario 4 — Summary Table

Defense	v1 Failed	v1 Accepted	Compromised	Networks	Pivoted
Baseline	60	18	8	3	YES
fail2ban	6	0	1	1	NO
Block IP	0	0	1	1	NO
Rate Limit	9	0	0	1	NO
Disable Pwd	0	0	1	1	NO
f2b + Rate	9	0	0	1	NO
All Defenses	0	0	1	1	NO

(d) S4

Fig. 7: Detailed per-host summary tables.